LEMMA PRESERVES COH

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter obsolete.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

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Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-preserves-Coh.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0G05: sources/varieties.tex, source lines 6073–6073. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G02: sources/equiv.tex, source lines 2531–2531. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G03: sources/equiv.tex, source lines 2578–2578. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 0BCD: sources/more-algebra.tex, source lines 19594–19594. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original reference 0EGV: sources/more-algebra.tex, source lines 28994–28994. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 0FXN: sources/more-algebra.tex, source lines 29816–29816. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G00: sources/equiv.tex, source lines 2772–2772. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0FZP: sources/functors.tex, source lines 2039–2039. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 08EU: sources/perfect.tex, source lines 5052–5052. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 039M: sources/etale.tex, source lines 1110–1110. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06NC: sources/descent.tex, source lines 6382–6382. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let k be a field. Let X be a separated scheme of finite type over k which is regular. Let $F : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_X)$ be a k-linear exact functor. Assume for every coherent $\mathcal{O}_X$-module $\mathcal{F}$ with $\dim(Supp(\mathcal{F})) = 0$ there is an isomorphism of k-vector spaces*

$$\mathrm{Hom}_X(\mathcal{F}, M) = \mathrm{Hom}_X(\mathcal{F}, F(M))$$

functorial in M in $D_{perf}(\mathcal{O}_X)$. Then there exists an automorphism $f : X \to X$ over k which induces the identity on the underlying topological space[1] and an invertible $\mathcal{O}_X$-module $\mathcal{L}$ such that F and $F'(M) = f^*M \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{L}$ are siblings.

Proof. By Derived Categories of Varieties, Lemma 0G02 we conclude that for every coherent $\mathcal{O}_X$-module $\mathcal{F}$ whose support is a closed point there are isomorphisms

$$H^0(X, M \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}) = H^0(X, F(M) \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F})$$

functorial in M.

[1] This often forces f to be the identity, see Varieties, Lemma 0G05.

Let $x \in X$ be a closed point and apply the above with $\mathcal{F} = \mathcal{O}_x$ the skyscraper sheaf with value $\kappa(x)$ at x. We find

$$\dim_{\kappa(x)} \operatorname{Tor}_p^{\mathcal{O}_{X,x}}(M_x, \kappa(x)) = \dim_{\kappa(x)} \operatorname{Tor}_p^{\mathcal{O}_{X,x}}(F(M)_x, \kappa(x))$$

for all $p \in \mathbf{Z}$. In particular, if $H^i(M) = 0$ for $i > 0$, then $H^i(F(M)) = 0$ for $i > 0$ by Derived Categories of Varieties, Lemma 0G03.

If $\mathcal{E}$ is locally free of rank r, then $F(\mathcal{E})$ is locally free of rank r. This is true because a perfect complex K over $\mathcal{O}_{X,x}$ with

$$\dim_{\kappa(x)} \operatorname{Tor}_i^{\mathcal{O}_{X,x}}(K, \kappa(x)) = \begin{cases} r & \text{if} \quad i = 0 \\ 0 & \text{if} \quad i \neq 0 \end{cases}$$

is equal to a free module of rank r placed in degree 0. See for example More on Algebra, Lemma 0BCD.

If M is supported on a closed subscheme $Z \subset X$, then $F(M)$ is also supported on Z. This is clear because we will have $M \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_x = 0$ for $x \notin Z$ and hence the same will be true for $F(M)$ and hence we get the conclusion from Derived Categories of Varieties, Lemma 0G03.

In particular $F(\mathcal{O}_x)$ is supported at $\{x\}$. Let $i \in \mathbf{Z}$ be the minimal integer such that $H^i(\mathcal{O}_x) \neq 0$. We know that $i \leq 0$. If $i < 0$, then there is a morphism $\mathcal{O}_x[-i] \to F(\mathcal{O}_x)$ which contradicts the fact that all morphisms $\mathcal{O}_x[-i] \to \mathcal{O}_x$ are zero. Thus $F(\mathcal{O}_x) = \mathcal{H}[0]$ where $\mathcal{H}$ is a skyscraper sheaf at x.

Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$-module with $\dim(\operatorname{Supp}(\mathcal{G})) = 0$. Then there exists a filtration

$$0 = \mathcal{G}_0 \subset \mathcal{G}_1 \subset \ldots \subset \mathcal{G}_n = \mathcal{G}$$

such that for $n \geq i \geq 1$ the quotient $\mathcal{G}_i/\mathcal{G}_{i-1}$ is isomorphic to $\mathcal{O}_{x_i}$ for some closed point $x_i \in X$. Then we get distinguished triangles

$$F(\mathcal{G}_{i-1}) \to F(\mathcal{G}_i) \to F(\mathcal{O}_{x_i})$$

and using induction we find that $F(\mathcal{G}_i)$ is a coherent sheaf placed in degree 0.

Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$-module. We know that $H^i(F(\mathcal{G})) = 0$ for $i > 0$. To get a contradiction assume that $H^i(F(\mathcal{G}))$ is nonzero for some $i < 0$. We choose i minimal with this property so that we have a morphism $H^i(F(\mathcal{G}))[-i] \to F(\mathcal{G})$ in $D_{perf}(\mathcal{O}_X)$. Choose a closed point $x \in X$ in the support of $H^i(F(\mathcal{G}))$. By More on Algebra, Lemma 0EGV there exists an $n > 0$ such that

$$H^i(F(\mathcal{G}))_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X,x}/\mathfrak{m}_x^n \longrightarrow \operatorname{Tor}_{-i}^{\mathcal{O}_{X,x}}(F(\mathcal{G})_x, \mathcal{O}_{X,x}/\mathfrak{m}_x^n)$$

is nonzero. Next, we take $m \geq 1$ and we consider the short exact sequence

$$0 \to \mathfrak{m}_x^m \mathcal{G} \to \mathcal{G} \to \mathcal{G}/\mathfrak{m}_x^m \mathcal{G} \to 0$$

By the above we know that $F(\mathcal{G}/\mathfrak{m}_x^m \mathcal{G})$ is a sheaf placed in degree 0. Hence $H^i(F(\mathfrak{m}_x^m \mathcal{G})) \to H^i(F(\mathcal{G}))$ is an isomorphism. Consider the commutative diagram

$$\begin{array}{ccc} H^i(F(\mathfrak{m}_x^m \mathcal{G}))_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X,x}/\mathfrak{m}_x^n & \longrightarrow & \operatorname{Tor}_{-i}^{\mathcal{O}_{X,x}}(F(\mathfrak{m}_x^m \mathcal{G})_x, \mathcal{O}_{X,x}/\mathfrak{m}_x^n) \\ \downarrow & & \downarrow \\ H^i(F(\mathcal{G}))_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X,x}/\mathfrak{m}_x^n & \longrightarrow & \operatorname{Tor}_{-i}^{\mathcal{O}_{X,x}}(F(\mathcal{G})_x, \mathcal{O}_{X,x}/\mathfrak{m}_x^n) \end{array}$$

Since the left vertical arrow is an isomorphism and the bottom arrow is nonzero, we conclude that the right vertical arrow is nonzero for all $m \geq 1$. On the other hand, by the first paragraph of the proof, we know this arrow is isomorphic to the arrow

$$\operatorname{Tor}_{-i}^{\mathcal{O}_{X,x}}(\mathfrak{m}_x^m \mathcal{G}_x, \mathcal{O}_{X,x}/\mathfrak{m}_x^n) \longrightarrow \operatorname{Tor}_{-i}^{\mathcal{O}_{X,x}}(\mathcal{G}_x, \mathcal{O}_{X,x}/\mathfrak{m}_x^n)$$

However, this arrow is zero for $m \gg n$ by More on Algebra, Lemma 0FXN which is the contradiction we're looking for.

Thus we know that F preserves coherent modules. By Derived Categories of Varieties, Lemma 0G00 we find F is a sibling to the Fourier-Mukai functor F' given by a coherent $\mathcal{O}_{X\times X}$-module $\mathcal{K}$ flat over X via pr_1 and finite over X via pr_2. Since $F(\mathcal{O}_X)$ is an invertible $\mathcal{O}_X$-module $\mathcal{L}$ placed in degree 0 we see that

$$\mathcal{L} \cong F(\mathcal{O}_X) \cong F'(\mathcal{O}_X) \cong \mathrm{pr}_{2,*}\mathcal{K}$$

Thus by Functors and Morphisms, Lemma 0FZP there is a morphism $s : X \to X \times X$ with $\mathrm{pr}_2 \circ s = \mathrm{id}_X$ such that $\mathcal{K} = s_*\mathcal{L}$. Set $f = \mathrm{pr}_1 \circ s$. Then we have

$$\begin{aligned} F'(M) &= R\mathrm{pr}_{2,*}(L\mathrm{pr}_1^*K \otimes \mathcal{K}) \\ &= R\mathrm{pr}_{2,*}(L\mathrm{pr}_1^*M \otimes s_*\mathcal{L}) \\ &= R\mathrm{pr}_{2,*}(Rs_*(Lf^*M \otimes \mathcal{L})) \\ &= Lf^*M \otimes \mathcal{L} \end{aligned}$$

where we have used Derived Categories of Schemes, Lemma 08EU in the third step. Since for all closed points $x \in X$ the module $F(\mathcal{O}_x)$ is supported at x, we see that f induces the identity on the underlying topological space of X. We still have to show that f is an isomorphism which we will do in the next paragraph.

Let $x \in X$ be a closed point. For $n \geq 1$ denote $\mathcal{O}_{x,n}$ the skyscaper sheaf at x with value $\mathcal{O}_{X,x}/\mathfrak{m}_x^n$. We have

$$\mathrm{Hom}_X(\mathcal{O}_{x,m}, \mathcal{O}_{x,n}) \cong \mathrm{Hom}_X(\mathcal{O}_{x,m}, F(\mathcal{O}_{x,n})) \cong \mathrm{Hom}_X(\mathcal{O}_{x,m}, f^*\mathcal{O}_{x,n} \otimes \mathcal{L})$$

functorially with respect to $\mathcal{O}_X$-module homomorphisms between the $\mathcal{O}_{x,n}$. (The first isomorphism exists by assumption and the second isomorphism because F and F' are siblings.) For $m \geq n$ we have $\mathcal{O}_{X,x}/\mathfrak{m}^n = \mathrm{Hom}_X(\mathcal{O}_{x,m}, \mathcal{O}_{x,n})$ via the action on $\mathcal{O}_{x,n}$ we conclude that $f^\sharp : \mathcal{O}_{X,x}/\mathfrak{m}_x^n \to \mathcal{O}_{X,x}/\mathfrak{m}_x^n$ is bijective for all n. Thus f induces isomorphisms on complete local rings at closed points and hence is étale (Étale Morphisms, Lemma 039M). Looking at closed points we see that $\Delta_f : X \to X \times_{f,X,f} X$ (which is an open immersion as f is étale) is bijective hence an isomorphism. Hence f is a monomorphism. Finally, we conclude f is an isomorphism as Descent, Lemma 06NC tells us it is an open immersion. $\square$